Analyzing earthquake clustering features by using stochastic reconstruction

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[1] On the basis of the epidemic-type aftershock sequence (ETAS) model and the thinning procedure, this paper gives the method about how to classify the earthquakes in a given catalogue into different clusters stochastically. The key points of this method are the probabilities of one event being triggered by another previous event and being a background event. Making use of these probabilities, we can reconstruct the functions associated with the characteristics of earthquake clusters to test a number of important hypotheses about the earthquake clustering phenomena. We applied this reconstruction method to the shallow seismic data in Japan and also to a simulated catalogue. The results show the following assertions: (1) The functions for each component in the formulation of the space-time ETAS model are good enough as a first-order approximation for describing earthquake clusters; (2) a background event triggers less offspring in expectation than a triggered event of the same magnitude; (3) the magnitude distribution of the triggered event depends on the magnitude of its direct ancestor; (4) the diffusion of the aftershock sequence is mainly caused by cascades of individual triggering processes, while no evidence shows that each individual triggering process is diffusive; and (5) the scale of the triggering region is still an exponential law, as formulated in the model but not the same one for the expected number of offspring. *INDEX TERMS:* 7209 Seismology: Earthquake dynamics and mechanics; 7223 Seismology: Seismic hazard assessment and prediction; 7260 Seismology: Theory and modeling; *KEYWORDS:* stochastic reconstruction, stochastic declustering, ETAS model, point process, aftershock, triggered seismicity

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1. Introduction

[2] Seismicity is clustered in both space and time. For the purpose of long-term earthquake prediction, such as zoning and earthquake potential estimation, researchers try to remove the temporary clustering to estimate the background seismicity; on the other hand, for the short-term or real-time prediction, we need a good understanding of the earthquake clusters. These are among the reasons why seismologists intensively study earthquake clusters. The Omori law is one of greatest successes in empirical studies [*Omori*, 1894] (and see *Utsu et al.* [1995] for a review). This empirical law has been generalized to statistical models or, more precisely, point process models, in different forms [*Ogata*, 1988, 1998; *Kagan*, 1991; *Musmeci and Vere-Jones*, 1992; *Rathbun*, 1993]. Among them, *Ogata* [1998] suggested a model with a component of Omori-type (inverse power law) decay in time and a component for the locations of triggered events, independent from the time component. In general, all of those models classify the seismicity into two components, the background and the cluster, where each earthquake event, no matter if it is a background event or generated by another event, produces (triggers) its own offspring (aftershocks) according to some branching rules.

[3] In the literature of seismology, people traditionally use some window-based methods [see, e.g., *Utsu*, 1969; *Gardner and Knopoff*, 1974; *Keilis-Borok and Kossobokov*, 1986] or link-based methods [e.g., *Reasenberg*, 1985; *Frohlich and Davis*, 1990; *Davis and Frohlich*, 1991] to decluster the catalogue or to identify earthquake clusters. The coefficients in these declustering rules depends to a large degree on the intuitions and experience of researchers. Alternative to these deterministic declustering methods, ideas of probability treatment to the background component and clustering component first were given by *Kagan and Knopoff* [1976]. *Zhuang et al.* [2002] suggested a method called stochastic declustering to bring such a probability treatment into practice. This method is based on stochastic models such as those mentioned above and can produce stochastic versions of declustered catalogues similar to the output of the conventional earthquake declustering methods. The core of the stochastic declustering method is an iterative approach to

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simultaneously estimate the background intensity, assumed to be a function of space but not of time, and the parameters associated with clustering structures. Making use of these estimates and the thinning operation, one can obtain the probabilities for each event being a background event or a triggered event. These probabilities are the key to realizing stochastic versions of the clustering family trees in the catalogue and, of course, to separating the background events from the earthquake clusters.

[4] Because these probabilities are estimated through a particular model, the closeness between the formulation of the model and the reality is the essentially important factor influencing the output. The closer the model is to the real data, the more reliable the output is. Of course, some model selection procedures can be used to choose the best model among many models fitted to the same set of data. However, these procedures usually give us a number indicating the overall goodness of fit for the model and rarely tell us whether there are some good points in a model if its overall fit is worse than another. It is also difficult to find clues about how to improve the formulation of the clustering models through model selection procedures. In this paper, we are going to show some graphical diagnostic methods useful for improving model formulation.

[5] In the earthquake catalogue, the background seismicity usually overlaps with the clusters, and the clusters may also overlap with each other, which complicate the test on the hypotheses about seismicity. However, as we show in this paper, the outputs from the stochastic declustering algorithm make it possible to evaluate the uncertainty or significance of some particular features associated with the background seismicity or the earthquake clusters.

[6] In sections 2 and 3, we will first give a brief description of the models on which we base the stochastic declustering method and outline some associated techniques, including variable kernel estimates, thinning procedures, and algorithms for estimating the cluster parameters and separating catalogue into earthquake family trees. Then, in sections 4–8, using the Japanese Meteorological Agency (JMA) catalogue as demonstration, we will explain how to use the stochastic declustering output to build empirical distribution functions to test a number of hypotheses associated with the earthquake clustering features. Some procedures will be also applied to a simulated catalogue for comparison.

2. Formulation of the ETAS Model and Stochastic Declustering

[7] Up to now, several branching models have been proposed for describing the space-time clustering features of earthquakes by *Kagan* [1991], *Rathbun* [1993], *Musmeci and Vere-Jones* [1992], *Ogata* [1998], and *Ogata et al.* [2003]. All these models can be formulated in the form of the conditional intensity function; that is, the process is controlled by an intensity conditional on the observation history (see, e.g., *Daley and Vere-Jones* [2002, Chapter 7] for details of mathematical treatment on the conditional intensity), namely,

$$\lambda(t,x,y,M|\mathcal{H}_t)dt\,dx\,dy\,dM = E[\mathbf{N}(dt\,dx\,dy\,dM)|\mathcal{H}_t], \quad (1)$$

where $\mathcal{H}_t$ is the observation history up to time t. In this study, we base our analysis on the formulation of the space-time epidemic-type aftershock sequence (ETAS) models by *Ogata* [1998],

$$\lambda(t,x,y,M|\mathcal{H}_t) = \lambda(t,x,y|\mathcal{H}_t)J(M) \quad (2)$$

$$\lambda(t,x,y|\mathcal{H}_t) = \mu(x,y) + \sum_{i:\,t_i<t} \kappa(M_i)g(t-t_i)f(x-x_i,y-y_i|M_i), \quad (3)$$

where (1) $\mu(x, y)$ is the background intensity, which is a function of space, but not of time; (2) $\kappa(M)$ is the expected number of events triggered from an event of a magnitude M, in the form

$$\kappa(M) = A\exp[\alpha(M-M_C)], \quad (4)$$

where A and α are constant and M_C is the magnitude threshold [see *Utsu*, 1969; *Yamanaka and Shimazaki*, 1990]; (3) $g(t)$ is the probability density function of the occurrence times of the triggered events, taking the form

$$g(t) = \frac{p-1}{c}\left(1+\frac{t}{c}\right)^{-p}, \quad (5)$$

i.e., the probability density function (pdf) form of the modified Omori law [*Utsu*, 1969]; (4) $f(x, y|M)$ is the location distribution of the triggered events, which is formulated in either of two ways, a short-range Gaussian decay and a long-range inverse power decay or, explicitly,

$$f(x,y|M) = \frac{1}{2\pi D^2 e^{\alpha(M-M_C)}}\exp\left[-\frac{x^2+y^2}{2D^2e^{\alpha(M-M_C)}}\right] \quad (6)$$

for the short-range decay and

$$f(x,y|M) = \frac{(q-1)D^{2(q-1)}e^{\alpha(q-1)(M-M_C)}}{\pi[x^2+y^2+D^2e^{\alpha(M-M_C)}]^q}, \quad q>1, \quad (7)$$

for the long-range decay; and (5) $J(M)$ is the probability density of magnitudes of all the events, independent from other components and taking the form of the Gutenberg-Richter law

$$J(M) = \beta e^{-\beta(M-M_c)}, \quad M \geq M_c, \quad (8)$$

where β is linked with the Gutenberg-Richter's b value by $\beta = b\log 10$ and M_c is the magnitude threshold considered.

[8] In the following, an ETAS model equipped with equation (6) is called model I and if equipped with equation (7) is called model II. The expected number of offspring that an event can trigger, namely, $\kappa(M)$, is also-called its triggering ability.

[9] In equations (4), (6), and (7), the spatial scaling factor for the direct aftershock region is proportional to the triggering ability of the ancestor. This judgment is from *Kanamori and Anderson* [1975], who tried to explain the exponential law for the number of aftershocks. In section 8.3, we will use the stochastic reconstruction method to verify whether it is a good choice.

More general discussions on using point processes to model seismicity are given by *Vere-Jones* [1992, 1995, 1999] and *Ogata* [1999].

2.1. Criticality

[10] The criticality of this process is determined by the critical parameter, ϱ, which is the biggest eigenvalue of

$$\varrho V(t,x,y) = \int_{-\infty}^{t} \iint_{\mathcal{R}^2} \int_{\mathcal{M}} \kappa(M^*) g(t-t^*) f(x-x^*, y-y^*) \cdot V(t^*, x^*, y^*) J(M^*) dM^* dx^* dy^* dt^*, \tag{9}$$

where $\mathcal{M}$ is the range of all the magnitudes. From equation (9), we find

$$\varrho = \int \kappa(M^*) J(M^*) dM^*. \tag{10}$$

Thus ϱ is the expected triggering ability of an event of arbitrary magnitude. When $\varrho < 1$, the process is stable (subcritical); when $\varrho \geq 1$, it is unstable. When the magnitude distribution satisfies the condition of assumption 5, this ϱ is also the proportion of all the triggered event in the total number of events. More detailed discussions on criticality are given by *Musmeci and Vere-Jones* [1992], *Zhuang* [2000, 2003], and *Helmstetter and Sornette* [2002, 2003].

2.2. Thinning Procedure

[11] One of the key points of the stochastic declustering method is the thinning operation on a point process determined by equation (3), which can split the whole process into several subprocesses [*Lewis and Shedler*, 1979; *Ogata*, 1981; *Daley and Vere-Jones*, 2002]. Suppose that events are numbered in time order from 1 to N. The probability of an event j being triggered by the ith event is

$$\rho_{ij} = \begin{cases} \dfrac{\zeta_i(t_j, x_j, y_j)}{\lambda(t_j, x_j, y_j | \mathcal{H}_{t_j})}, & j > i, \\ 0, & \text{otherwise}, \end{cases} \tag{11}$$

where

$$\zeta_i(t,x,y) = \kappa(M_i) g(t-t_i) f(x-x_i, y-y_i | M_i) \tag{12}$$

represents the intensity triggered by the ith event. Moreover, the probability of the event j being a triggered event is

$$\rho_j = \sum_{i<j} \rho_{ij}, \tag{13}$$

and the probability that the jth event belongs to the background is

$$\varphi_j = 1 - \rho_j = \frac{\mu(x_j, y_j)}{\lambda(t_j, x_j, y_j | \mathcal{H}_{t_j})}. \tag{14}$$

If we select each event j with probabilities ρ_{ij}, ρ_j, or φ_j, we can form up a new process being the triggered process by the ith event, the clustering process, or the background process, respectively.

2.3. Variable Kernel Estimates

[12] Once a background process is obtained, we can estimate the background intensity by some smoothing techniques. Rather than repeat the thinning procedure and the kernel estimation procedure many times to get an average estimate of the background intensity, we directly estimate the average by weighting, i.e.,

$$\hat{\mu}(x,y) = \frac{1}{T} \sum_i \varphi_i Z_{h_i}(x - x_i, y - y_i), \tag{15}$$

where i runs over all of the events in the whole process, T is the length of the time period of the process, and Z_h is the Gaussian Kernel function with a bandwidth h. The variable bandwidth h_j is determined by

$$h_j = \max\{\epsilon, \inf(r : \mathbf{N}[B(x_i, y_i; r)] > n_p)\}, \tag{16}$$

where ϵ is a small real number, $B(x, y; r)$ is the disk centered at (x, y) with a radius of r, and n_p is positive integer, i.e., h_j is the distance to n_pth closest event. Similar locally dependent estimates are also given by *Choi and Hall* [1999], *Musmeci and Vere-Jones* [1986], and *Silverman* [1986].

2.4. Maximum Likelihood Estimates

[13] Given an estimated intensity function $u(x, y)$, we set

$$\mu(x,y) = \nu u(x,y) \tag{17}$$

for the background rate, where ν is a positive value parameter, and then obtain the maximum likelihood estimates (MLE), $\hat{\theta} = (\hat{\nu}, \hat{A}, \hat{\alpha}, \hat{c}, \hat{p}, \hat{D})$ for mode I and $\hat{\theta} = (\hat{\nu}, \hat{A}, \hat{\alpha}, \hat{c}, \hat{p}, \hat{D}, \hat{q})$ for mode II, by maximizing the log likelihood

$$\log L(\theta) = \sum_k \log \lambda_\theta(t_k, x_k, y_k | \mathcal{H}_{t_k}) - \int_0^T \iint_S \lambda_\theta(t,x,y|\mathcal{H}_t) dx dy dt, \tag{18}$$

where k runs over all the events in the study regions S and the study time interval $[0, T]$ (not necessarily all the events observed). Thus we can estimate the background seismicity and the parameters in the clustering structures simultaneously by an iterative approach as outlined in Algorithm A as follows [*Zhuang et al.*, 2002].

[14] Algorithm A is an estimation of the parameters and the background intensity of the space-time ETAS model in the following steps.

In step A1, given a fixed n_p and ϵ, say 5 and 0.05 degrees (equivalent to 5.56 km on Earth's surface, which is close to the location error of earthquakes), calculate the bandwidth h_j for each event (t_j, x_j, y_j, M_j), $j = 1, 2, \cdots, N$.

In step A2, set $\ell = 0$ and $u^{(0)}(x, y) = 1$.

In step A3, using the maximum likelihood procedure [see, e.g., *Ogata*, 1998], fit the model with conditional intensity function

$$\lambda(t,x,y|\mathcal{H}_t) = \nu u^{(\ell)}(x,y) + \sum_{i:\, t_i < t} \kappa(M_i) g(t-t_i) \cdot f(x-x_i, y-y_i | M_i) \tag{19}$$

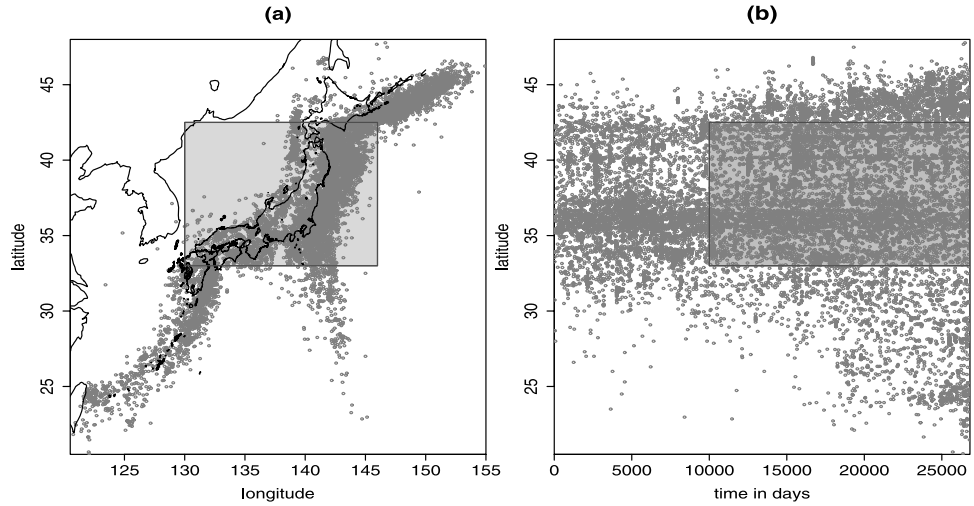


Figure 1. Seismicity in the Japan region and nearby during 1926–1999 ($M_J \geq 4.2$). (a) Epicenter locations. (b) Latitudes of epicenter locations versus occurrence times. The shaded region represents the study space-time range.

to the earthquake data, where κ, g, and f are defined in equation (2).

In step A4, calculate ρ_{ij}, ρ_j, and φ_j for $j = 1, 2, \ldots, N$ and each $i < j$ using equations (11), (13), and (14).

In step A5, calculate $\mu(x, y)$ by using equation (15), and record it as $u^{(\ell+1)}(x, y)$.

In step A6, if max $|u^{(\ell+1)}(x, y) - u^{(\ell)}(x, y)| > \varepsilon$, where ε is a given small positive number, then set $\ell = \ell + 1$ and go to step A3; otherwise, take $\nu u^{(\ell+1)}(x, y)$ as the background rate and also output ρ_{ij}, ρ_i, and φ_i.

3. Stochastic Reconstruction

[15] Algorithm A outputs the probabilities, φ_i and ρ_{ij}, which can be used to classify events in the catalogue into family trees. The algorithm for such a classification can be implemented as follows.

Algorithm B is the stochastic classification of earthquake clusters in the following steps.

In step B1, calculate φ_j and ρ_{ij} by using algorithm A, where $j = 1, 2, \ldots, N$ and $i = 1, 2, \ldots, j$, N being the total number of events.

In step B2, generate a random variable U_j uniformly distributed on [0, 1].

In step B3, for each j, if $U_j < \varphi_j$, then select j as a background or initial event; else, select I_j to be the parent of j, where

$$I_j = \min\left\{k : \varphi_j + \sum_{i=1}^{k} \rho_{ij} \geq U_j, \quad 1 \leq k < j\right\}.$$

[16] If we select the initial event in each family tree, we obtain a declustered catalogue containing the background events. The declustered catalogue produced in this way is different from those by using the conventional declustering methods, for the events selected here are always the earliest events in the family trees, but not the largest events. So the so-called "main shocks" in the conventional literature may not be included in the declustered catalogue. However, if preferred, we can select the largest events in each cluster as the representatives of the background seismicity.

[17] Algorithm B tackles the difficulties in testing hypotheses associated with earthquake clustering features, which are caused by the complicated overlapping of the background seismicity and different earthquake clusters in both space and time. We can repeat algorithm B many times to get different stochastic versions of separations of the earthquake clusters. The nonuniqueness of such realizations illustrates the uncertainty in determining earthquake clusters, and thus repetition can help us to evaluate the significance of some properties of seismicity clustering patterns. However, we can also implement these tests by working with the probabilities φ_j and ρ_{ij} directly. In section 4–8, we will show how to use these probabilities to reconstruct the characteristics associated with earthquake clustering features, using the JMA catalogue as an example. The same reconstruction procedures will be also applied to a simulated catalogue for comparison.

4. Data Description and Preliminary Results

[18] We use the JMA catalogue for the analysis. The selected data set is in the ranges of longitude 121°–155°E, latitude 21°–48°N, depth 0–100 km, time 1 January 1926 to 31 December 1999, and magnitude $\geq M_J$4.2. There are 19,139 events in this data set.

[19] As with other earthquake catalogues that cover a long history, completeness and homogeneity are always problems, causing trouble for statistical analysis. The incompleteness of the early period and inhomogeneity in this data set can be easily seen from Figure 1. To tackle these problems, we choose a study space-time range in which the seismicity seems to be relatively complete and homogeneous, with ranges of longitude 130°–146°E, latitude 33°–42.5°N, a time period of 10,000–26,814 days after 1 January 1926, and the same depth and magnitude

Table 1. List of Parameters Estimated From Fitting the ETAS Models to the JMA Catalogue[a]

Model	A, events/day	α, M^{-1}	c, day	p	D^2, degree^2	q	log L	ϱ	ϱ'	ϱ''
I	0.1909	1.365	0.01726	1.089	1.414×10^{-3}	na	−45,658	0.4379	0.5460	0.4431
II	0.1977	1.334	0.01903	1.102	8.663×10^{-4}	1.691	−45,068	0.4219	0.5454	0.4609

[a]The criticality parameter ϱ is estimated by substituting equations (8)–(10), i.e., $\varrho = A\beta/(\beta - \alpha)$; ϱ' is another estimate of the criticality parameter, by taking the average of the values of $\kappa(M)$ over all the events in the study time interval and the study regions; ϱ'' is the proportion of triggered events in the total events.

ranges as the whole data set. There are 8283 events occurring in the study space-time range.

[20] The earthquake events not in the target region and time period but in the data set, namely, the boundary catalogue, are used as the boundary effects in the estimation of the clustering parameters and the background rate in the model; that is, these events are not counted in computing the likelihood function (18) but are put in the observation history when evaluating the conditional intensity equation (3).

[21] We fitted both models I and II to this data set by using algorithm A. The results are outlined in Table 1.

[22] The background intensity of the study region is shown in Figure 2. As a feature complementary to the background, we define

$$\omega(x,y) = 1 - \frac{\mu(x,y)}{\Lambda(x,y)} \tag{20}$$

as the clustering coefficient to measure the clustering effect relative to the total intensity at a location (x, y). This function is as important as $\mu(x, y)$ in earthquake hazard estimation because damages caused by aftershocks cannot be ignored. An example of separating the background seismicity and the clustering seismicity is shown in Figure 3, obtained by using the stochastic declustering algorithm.

5. Simulation

[23] Simulation is a quite important tool in understanding the ETAS model because it can help us to visualize what the model describes and also to find out the discrepancy between the real catalogue and the model. *Ogata* [1998] used the thinning method for simulating synthetic catalogues. Here we use a method based on the branching structures of the ETAS model [*Kagan*, 1991; R. Davies, personal communications, 1997].

[24] Algorithm C is a simulation of the space-time ETAS model using the following steps.

In step C1, generate the background catalogue with the estimated background intensity $\mu(x, y)$ in equation (3), recorded as generation 0, namely, $G^{(0)}$, according to the simulation for the nonhomogeneous stationary space-time Poisson process [see, e.g., *Daley and Vere-Jones*, 2002, section 7.4].

In step C2, set $\ell = 0$.

In step C3, for each event i, namely, (t_i, x_i, y_i, M_i), in the catalogue $G^{(\ell)}$, simulate its $N^{(i)}$ offspring, namely, $O_i^{(\ell)} = \{(t_k^{(i)}, x_k^{(i)}, y_k^{(i)}, M_k^{(i)}): k = 1,\ldots, N^{(i)}\}$, where $N^{(i)}$ is a Poisson random variable with a mean of $\kappa(M_i)$, and $t_k^{(i)}$, $(x_k^{(i)}, y_k^{(i)})$, and $M_k^{(i)}$ are generated from the probability densities $g(t - t_i)$, $f(x - x_i, y - y_i|M_i)$, and $J(M)$, respectively.

In step C4, set $G^{(\ell+1)} = \bigcup_{i\in G^{(\ell)}} O_i^{(\ell)}$.

In step C5, if $G^{(\ell)}$ is not empty, let $\ell = \ell + 1$ and goto step C3; else return $\bigcup_{j=0}^{\ell} G^{(j)}$.

[25] Algorithm C gives the procedure for simulating a point process controlled by the conditional intensity of the ETAS model in principle. For the simulation in this study, we consider the following points to take the advantage of having the original JMA catalogue, to which the model is fitted.

[26] The magnitudes of the events can be generated by using the Gutenberg-Richter law, i.e., the exponential distribution with a density of $J(M) = \beta e^{-\beta(M-M_C)}$. In this way, the largest magnitude, $M_{(n)}$, in n samples has the expectation,

$$E\left[M_{(n)}\right] = \frac{1}{\beta}\sum_{i=1}^{n}\frac{1}{i} + M_C \sim \frac{\log n + C_E}{\beta} + M_C,$$

where $C_E = \lim_{n\to\infty}[\sum_{i=1}^{n} 1/i - \log n] \approx 0.5772\cdots$ is the Euler constant. In this study, there are $N = 8283$ events in

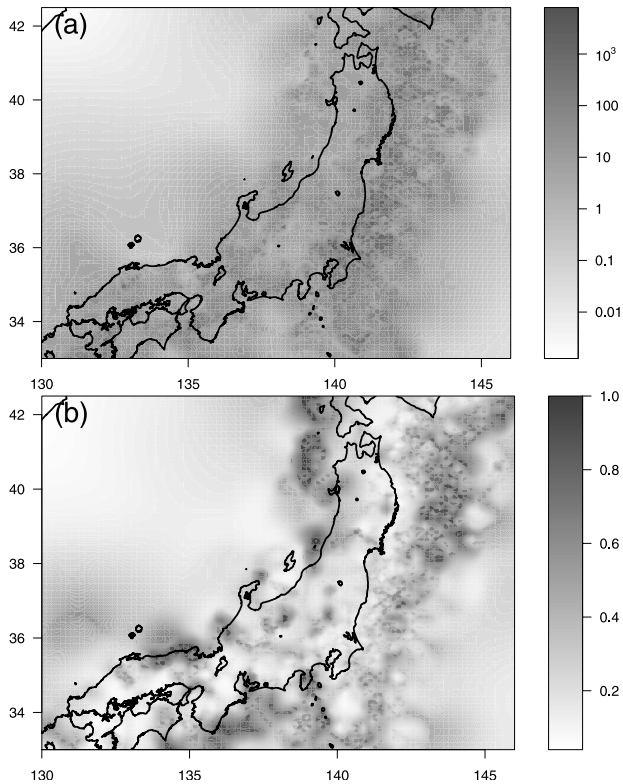


Figure 2. (a) Estimated background rate $\hat{\mu}$ in equation (15) in the study region (unit is events/(degree^2 × 74 years)). (b) Clustering coefficient in equation (20).

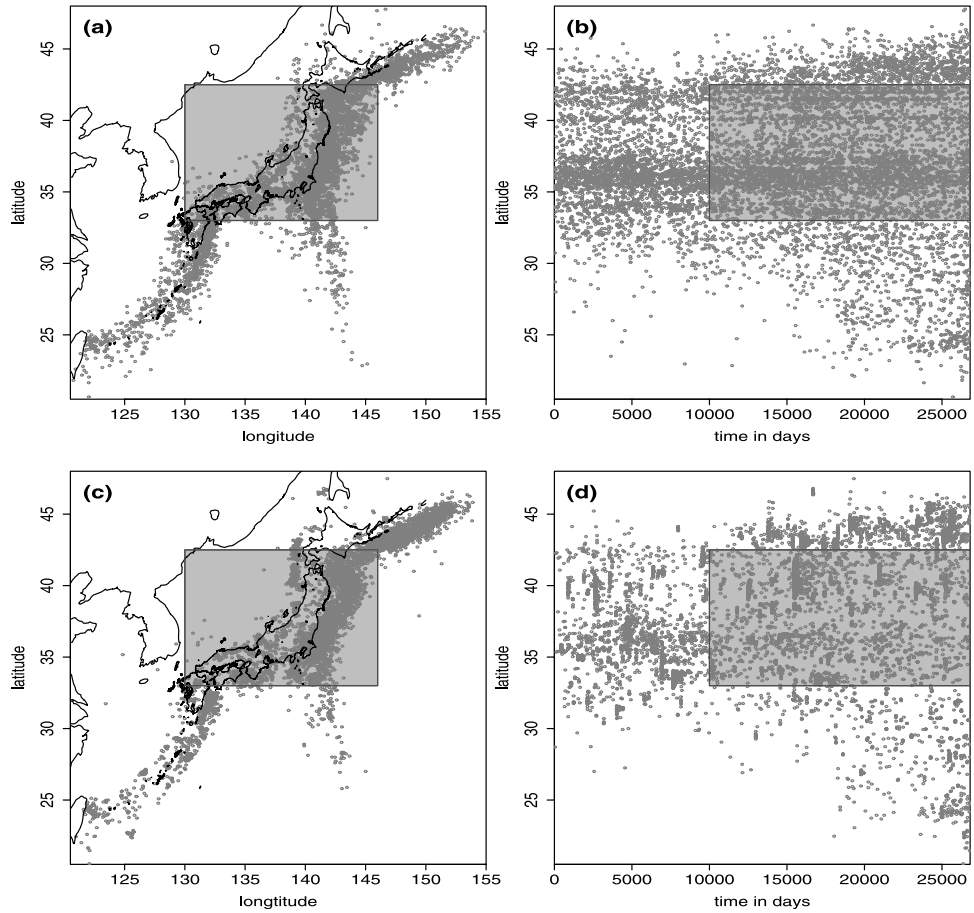


Figure 3. Space-time plots for a version of stochastically declustered catalogue by applying algorithm B to the outputs from fitting model II to the JMA data. The shaded rectangle represents the study space-time range. (a) Epicenter locations of the background events. (b) Latitudes versus occurrence times of the background events. (c) Epicenter locations of the triggered events. (b) Latitudes versus occurrence times of the triggered events.

the JMA catalogue falling in the target space-time ranges, indicating that $E[M_{(N)}] = 9.101$, about 0.9 more than the maximum magnitude, 8.2, in the catalogue. To avoid generating so large a magnitude, we can use some other distributions to replace the exponential one, such as the tapered exponential distribution suggested by *Kagan and Schoenberg* [2001], to replace equation (8) in the model formulation. However, in this paper, the magnitudes of the simulated events can be resampled from magnitudes of the events falling within the study space-time range in the JMA catalogue.

[27] It is not easy to simulate the background process with an intensity function $\mu(x, y)$. In this study, we can generate a background catalogue by applying algorithm B to the JMA catalogue. We then add a Gaussian deviation, with a zero mean and a standard error of h_i defined in equation (16), to the location of each selected background event i. This step helps us to produce a background catalogue whose spatial occurrence rate is exactly as the variable kernel estimate $\nu\hat{\mu}(x, y)$ in equation (15). To obtain the independence feature between occurrence times and locations in the background seismicity, we then keep the order of occurrence times but randomly reorder the epicenter locations. One of the advantages of taking such a random permutation is that even if the background seismicity in the original JMA catalogue is not stationary, the simulated background catalogue minimizes the statistical differences between the JMA catalogue and the simulated catalogue, brought by nonstationarity. Finally, the magnitude of each event in the background catalogue is resampled from the collection of the magnitudes of all the events falling within the study space-time range in the JMA catalogue.

[28] Boundary effect can never be neglected, for the simulation of a space-time ETAS is always carried out in a given region and a given time period. If the background catalogue is simulated only inside this target space-time range, the influence from the events in earlier history or outside the target region is neglected, and a fewer number of triggered events are produced. To avoid such boundary effect, we need carry out the simulation within a larger space-time range and then select events in the target space-time range to form up the simulated catalogue. However, in this study, since the boundary catalogue can be taken as the events of the JMA catalogue falling out of the study space-time range, we alternatively can run the simulation in the study space-time range with the union of the background catalogue and the simulated background catalog as generation 0, $G^{(0)}$, to simulated generation 1. To

avoid counting boundary effect twice, once a new generation is simulated, the new events outside the study space-time range are removed before simulating the next generation. The simulated catalogue is shown in Figure 4.

6. Testing Individual Functions in the Model Formulation

[29] Needless to say, a successful choice of the model formulation is essential for a good fit to the data. To justify the particular models used in this study, we can use likelihood-based model selection procedures and apply the residual analysis [see, e.g., *Schoenberg*, 2003] to find the overall goodness of fit of the model to the data. These routines compare the overall fittings between different models, and rarely tell us the goodness of fit for each component in the branching structure of the ETAS model. In this study, we will use the stochastic reconstruction method to test whether the features in the model formulation are also true for the earthquake data. In sections 6–8, first, we will check whether the choice of the formulation of $\kappa(M)$, $g(t)$, and $f(x, y|M)$ are reasonable; then we will compare the characteristics between the background events and the triggered events; and finally, we will test the existence of dependence between some clustering components.

6.1. Triggering Abilities

[30] For each integer pair (i, j) such that $0 < i < j \leq N$, where N is the number of total events, the jth event can be selected as the direct offspring of the ith event with probability ρ_{ij}. Furthermore, the expected number of events in a realization of the subprocess triggered by the ith event, by using algorithm B, is

$$S_i = \sum_j \rho_{ij}. \tag{21}$$

Thus the expected number of offspring triggered by an event of magnitude M can be estimated through

$$\hat{\kappa}(M) = \frac{\sum_i S_i}{\sum_i I(|M_i - M| < \Delta M/2)} = \frac{\sum_i \sum_j \rho_{ij}}{\sum_i I(|M_i - M| < \Delta M/2)}, \tag{22}$$

where ΔM is a small positive number, and I is the index function such that

$$I(x) = \begin{cases} 1, \text{if the logical statement x holds,} \\ 0, \text{else.} \end{cases} \tag{23}$$

Because magnitudes in most catalogues are rounded to 0.1, we can simply take $\Delta M = 0.05$ to get a discrete form for $\hat{\kappa}(M)$.

[31] The reconstruction results of $\hat{\kappa}(M)$ are shown in Figure 5. Roughly speaking, the exponential law fits $\kappa(M)$ quit well overall, indicating that it can be used as the first-step approximation for the triggering law. However, there are also some clear discrepancies between the formulated functions and the reconstructed ones: the reconstructed curves are lower than the theoretical ones between the magnitude range from $M_J5.5$ to M_J7, and higher for $M_J7.2$. For the simulated catalogue, the differences between the reconstructed $\hat{\kappa}(M)$ and the theoretical $\kappa(M)$ are very small and can be ignored.

6.2. Time Lag Distributions

[32] We rebuild the probability density function of the time differences between the ancestors and their direct offspring by

$$\hat{g}(t) = \frac{\sum_{i,j} \rho_{ij} I\left(\left|t_j - t_i - t\right| < \Delta t/2\right)}{\Delta t \sum_{i,j} \rho_{i,j}}, \tag{24}$$

where Δt is a small number, as shown in Figure 6. The main discrepancies between the theoretical function and the reconstructed function are at the two ends of the curves, where the time difference is less than 0.1 day or greater than 5000 days. It is clear that the dropdown at the end of the large time lags is due to the boundary effect produced by the absence of the events occurring after or before the observation period. Even though this curvature could be corrected, in principle, we are not going to do so in order not to complicate the analysis. The dropdown at the beginning of the reconstructed curves is probably caused by the low detection rate of smaller events occurring immediately after a large earthquake because the seismic waves that they produce are mixed up on the seismographs. The agreement between the theoretical curves and the corresponding reconstructed functions in the large range of time lags from 0.1 to 5000 days sufficiently shows a good fit of the Omori law to the occurrence times of triggered events.

6.3. Location Distributions

[33] Define the standardized distance between a triggered event j and its direct ancestor, assumed i, by

$$r_{ij} = \sqrt{\frac{\left(x_j - x_i\right)^2 + \left(y_j - y_i\right)^2}{D^2 \exp[\alpha(M_i - M_c)]}}. \tag{25}$$

From equations (6) and (7), r_{ij} has a density function of

$$f_R(r) = 2re^{-r^2},\ r \geq 0, \tag{26}$$

for model I and

$$f_R(r) = \frac{2r(q-1)}{(1+r^2)^q}, r \geq 0, \tag{27}$$

for model II, respectively. The distribution with a density of equation (26) is called a Rayleigh distribution. On the other hand, $f_R(r)$ can be reconstructed through

$$\hat{f}_R(r) = \frac{\sum_{i,j} \rho_{ij} I\left(\left|r_{ij} - r\right| < \Delta r/2\right)}{\Delta r \sum_{i,j} \rho_{ij}}, \tag{28}$$

where Δr is a small positive number. The comparison between $\hat{f}_R$ and f_R for the two models is shown in Figure 7. It can be seen that if model I is used, the reconstructed

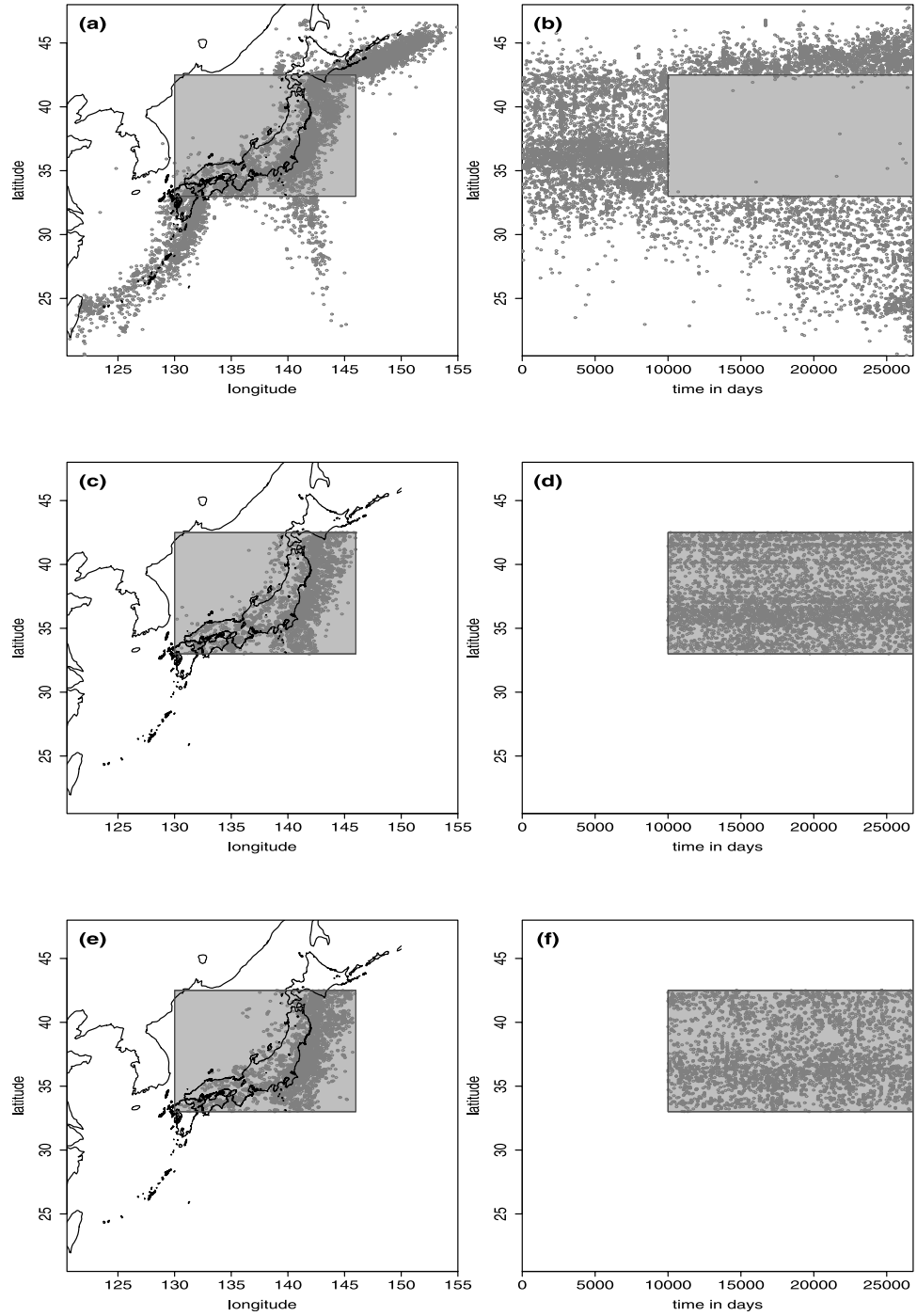


Figure 4. Synthetic catalogue simulated by using algorithm C with model II and parameters from fitting model II to the JMA catalogue. (a) Epicenter locations and (b) latitudes versus occurrence times of the events in the boundary catalogue, i.e., events not in the study space-time region in the JMA catalogue. (c) Epicenter locations and (d) latitudes versus occurrence times of the simulated background catalogue, generating by randomly arranging the order of the locations of a background catalogue produced by algorithm B and also resampling the magnitude from these of the events in the study space-time range in the JMA catalogue. (e) Epicenter locations and (f) latitudes versus occurrence times of the simulated triggered events, produced by using steps C2 to C5 of algorithm C.

probability density of the transformed distances between the ancestors and the direct offspring is quite different from the theoretical one. When model II is used, the reconstructed probability density is very close to the theoretical one. These results confirm that the aftershocks decay in a long range in space rather than in a short range [*Ogata*, 1998; *Console and Murru*, 2001; *Console et al.*, 2003]. These results also imply the robustness of the reconstruction

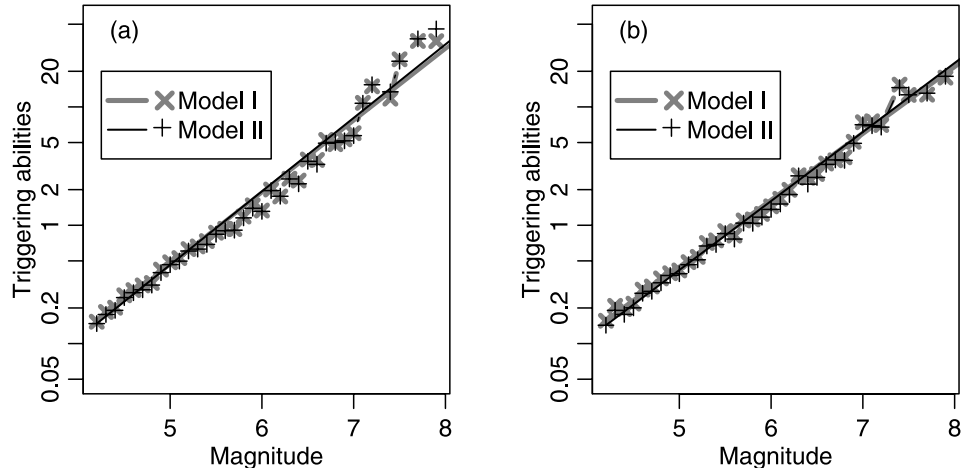


Figure 5. Reconstruction results for the triggering abilities $\hat{\kappa}(M)$ in equation (22) for (a) the JMA catalogue and (b) the simulated catalogue. Theoretical curves, $\kappa(M) = Ae^{\alpha(M-M_C)}$, are plotted by solid lines, where A and α are the MLE from fitting the models to the catalogues.

method, for we can get a reconstructed probability density function very close to the corresponding function in model II, even if an improper model like model I is employed. Since model II fits the seismicity much better than model I, we only consider model II for reconstruction in sections 7 and 8.

7. Comparison Between Features of Background Events and Triggered Events

[34] One of the most important assumptions of the space-time ETAS model is that there is no distinction between the main shocks and the aftershocks. Once an event occurs, no matter if it is background or triggered by a previous event, its magnitude is drawn from the unique magnitude distribution for all events, and it triggers offspring in the same manner as all the other events. In this section, we will verify whether there exist differences in the magnitude distributions, triggering abilities, and distributions of occurrence times and locations of the offspring between the background events and the triggered events.

7.1. Magnitude Distribution

[35] The empirical probability density functions of the magnitudes for the background events, the triggered events, and all the events can be reconstructed through

$$\hat{J}_b(M) = \frac{\sum_i \varphi_i I(|M_i - M| < \Delta M/2)}{\Delta M \sum_i \varphi_i}, \tag{29}$$

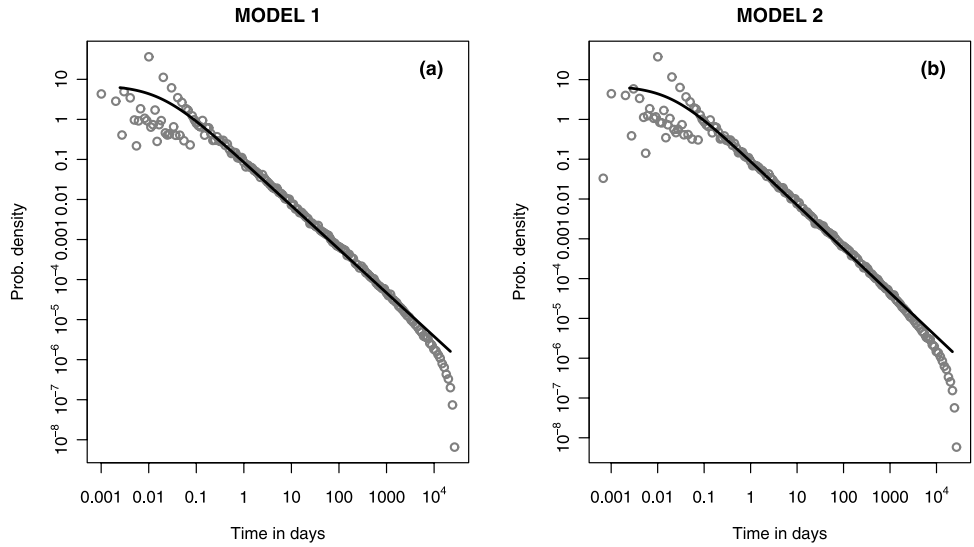


Figure 6. Reconstruction results of the time lag distribution $\hat{g}(t)$ in equation (24) for the JMA catalogue by using (a) model I and (b) model II. Theoretical curves of $g(t)$ in equation (5) are plotted by solid lines, with the parameters of the MLEs from fitting the models to the catalogue.

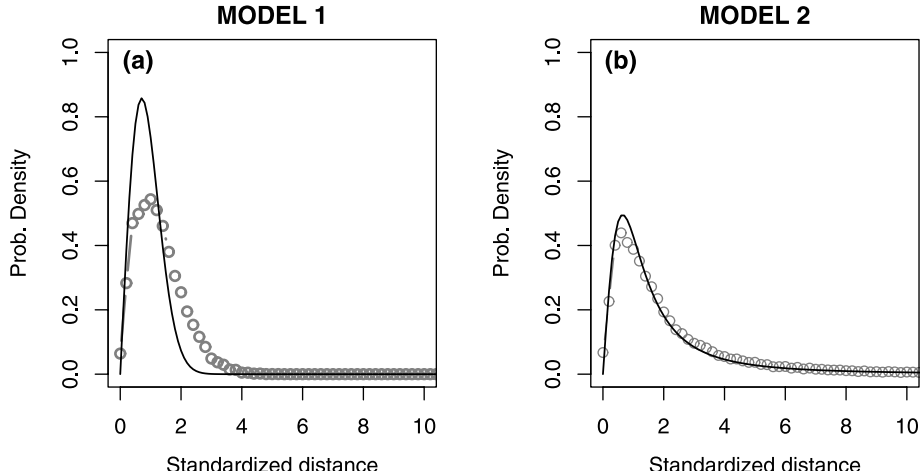


Figure 7. Reconstruction results for the distribution of the standardized triggering distances $\hat{f}_R(r)$ in equation (28) (gray circles) by using model I (a) and model II (b). The theoretical curves of $f_R(r)$ in equations (26) and (27) are plotted in solid lines in Figures 7a and 7b, respectively.

$$\hat{J}_t(M) = \frac{\sum_i (1-\varphi_i) I(|M_i - M| < \Delta M/2)}{\Delta M \sum_i (1-\varphi_i)}, \tag{30}$$

and

$$\hat{J}(M) = \frac{\sum_i I(|M_i - M| < \Delta M/2)}{\Delta M \sum_i 1}, \tag{31}$$

respectively, where ΔM is a small positive number. We use all the events in the target region and time period to reconstruct $\hat{J}_b(M)$ and $\hat{J}_t(M)$ for both the JMA catalogue and the simulated catalogue, as shown in Figure 8. The results show that for the JMA catalogue, the probability of a small event is higher for the background events than for the triggered events. Such a phenomenon cannot be found in the results for the simulated catalogue.

[36] If we assume that $J_b(M)$, $J_t(M)$, and $J(M)$ are all exponential distribution with parameters β_b, β_t, and β, respectively (strictly, if $J_b(M)$ and $J_t(M)$ are both exponential distributions with different parameters, then $J(M)$ is not exponential any more but is a mixture of two exponential densities). These parameters can be estimated through maximizing the following weighted likelihood functions:

$$\hat{\beta}_b = \frac{\sum_i \varphi_i}{\sum_i \varphi_i (M_i - M_c)}, \tag{32}$$

$$\hat{\beta}_t = \frac{\sum_i (1-\varphi_i)}{\sum_i (1-\varphi_i)(M_i - M_c)}, \tag{33}$$

$$\hat{\beta} = \frac{\sum_i 1}{\sum_i (M_i - M_c)}, \tag{34}$$

where i runs over all the events within the study space-time range.

[37] Table 2 shows the difference between the β values of the background events and the triggered events in the target region. It is possible that the estimates of these β values are influenced by the incompleteness for the low magnitudes, as shown in Figures 8c and 8d. However, such incompleteness cannot explain the differences between β values of different types of events in the JMA catalogue. To test the magnitude distributions of the background events and the triggered events, we carry out the Kalmogorov-Smirnov test and find that the hypothesis $J_b(M) = J_t(M)$ is clearly rejected.

[38] The β value for the background events is larger than that for the triggered events. This seems to be contrary to our knowledge that main shocks usually have smaller b values than the aftershocks [*Utsu*, 1969]. In fact, the higher b value for the main shocks is artificial in some sense. Because the main shocks are conventionally the largest events in the clusters identified by using some window-based or link-based declustering methods and such a selection increases the mean magnitude, a smaller b value is obtained. The background events here are the initial events of each cluster, and thus such a higher b value (β value) indicates that the clusters tend to be initiated by small events.

7.2. Triggering Abilities

[39] The triggering abilities of the background events and the triggered events for the JMA catalogue and the simulated catalogue can be reconstructed by using

$$\hat{\kappa}_b(M) = \frac{\sum_i \sum_j \varphi_i \rho_{ij} I(|M_i - M| < \Delta M/2)}{\sum_i \varphi_i I(|M_i - M| < \Delta M/2)} \tag{35}$$

and

$$\hat{\kappa}_t(M) = \frac{\sum_i \sum_j (1-\varphi_i) \rho_{ij} I(|M_i - M| < \Delta M/2)}{\sum_i (1-\varphi_i) I(|M_i - M| < \Delta M/2)}, \tag{36}$$

respectively, as shown in Figure 9. Both the background events and the triggered events generate offspring approxi-

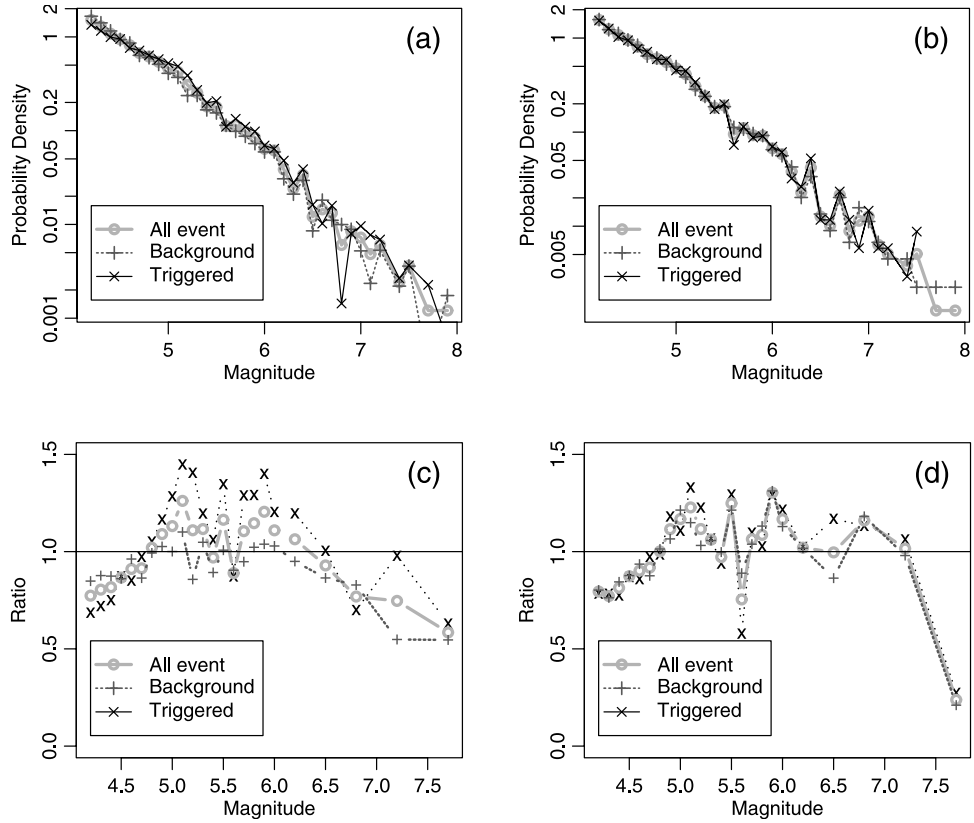


Figure 8. Reconstruction of the magnitude distributions, $\hat{J}_b(M)$ of the background events, $\hat{J}_t(M)$ of the triggered events, and $\hat{J}(M)$ of all the events (a) for the JMA catalogue and (b) for the simulated catalogue. Corresponding ratios of the reconstructed densities to the theoretic density, $\hat{J}_b(M)/J(M)$, $\hat{J}_t(M)/J(M)$, $\hat{J}(M)/J(M)$ for (c) the JMA catalogue and (d) the simulated catalogue, where $J(M)$ is defined in equation (8).

mately according to different exponential laws. For the same ancestor magnitude, a triggered event generates more offspring than a background event. The higher the magnitude, the smaller the difference. The results for the simulated catalogue show that there is no difference in triggering abilities between these two types of events, indicating that these differences are not caused by numerical procedures.

[40] In the ETAS model, the background events and triggered events generate offspring in the same way. The reconstruction results show that we should have different exponential laws for the triggering abilities of these two types of event, which is a possible direction on improving the current ETAS models, i.e., to treat the background events and triggered events separately in the model. The higher triggering abilities of the triggered events may be because they occur in an environment where the stress field is adjusting to the stress changes caused by their ancestors.

[41] We should notice that there seems to be a change point of slopes between $M6.0$ and $M7.0$ in the curves of triggering abilities. Such change points have been also reported by *Shimazaki* [1986], *Pacheco et al.* [1992], *Ikeya and Huang* [1997], and *Legrand* [2002] in the b values of the Gutenberg-Richter law or fractal dimension and are explained as being related to the thickness of the crust or seismogenic layer. The changes in the slope of the triggering ability may indicate that the total number of aftershocks is related to the geometry of the main shock fault.

7.3. Time Lags and Locations

[42] The distribution of the time lags and distances between a background event and its direct offspring can be estimated through

$$\hat{g}_b(t) = \frac{\sum_{i,j} \varphi_i \rho_{ij} I\left(\left|t_j - t_i - t\right| < \Delta t/2\right)}{\Delta t \sum_{i,j} \varphi_i \rho_{ij}} \tag{37}$$

Table 2. The β Values of Magnitude Distributions of Background and Clusters, Estimated by Using Equations (32)–(34)[a]

	Model I	Model II
	JMA Catalogue	
All events	1.9585 ± 0.0215	1.9585 ± 0.0215
Background	2.0964 ± 0.0309	2.1015 ± 0.0309
Triggered	1.8090 ± 0.0299	1.8141 ± 0.0298
	Simulated Catalogue	
All events	1.9537 ± 0.0203	1.9537 ± 0.0203
Background	1.9583 ± 0.0296	1.9551 ± 0.0295
Triggered	1.9501 ± 0.0282	1.9515 ± 0.0280

[a]See equation (8) for β values.

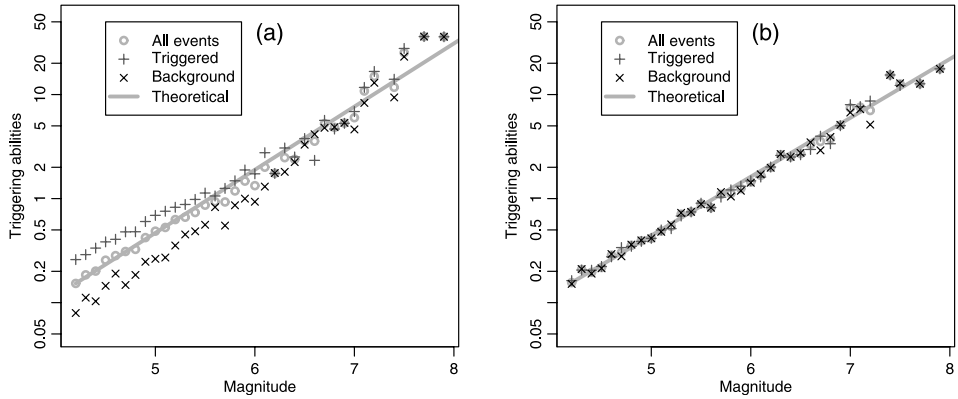


Figure 9. Reconstruction of the triggering abilities, $\hat{\kappa}_b(M)$ in equation (35) of the background events and $\hat{\kappa}_t(M)$ in equation (36) of the triggered events for (a) the JMA catalogue and (b) the simulated catalogue. For comparison, the empirical functions of the triggering abilities, $\hat{\kappa}(M)$ in equation (22) for all the events are plotted as gray circles, and the corresponding theoretical functions, $\kappa(M) = Ae^{\alpha(M-M_C)}$, are represented by the straight lines.

$$\hat{f}_{Rb}(r) = \frac{\sum_{i,j} \varphi_i \rho_{ij} I\left(\left|r_{ij} - r\right| < \Delta r/2\right)}{\Delta r \sum_{i,j} \varphi_i \rho_{ij}}. \tag{38}$$

Replacing the subscript b by t and φ by $1 - \varphi$, we can get the corresponding empirical probability density for the nonbackground ancestors, i.e.,

$$\hat{g}_t(t) = \frac{\sum_{i,j} (1 - \varphi_i) \rho_{ij} I\left(\left|t_j - t_i - t\right| < \Delta t/2\right)}{\Delta t \sum_{i,j} (1 - \varphi_i) \rho_{ij}} \tag{39}$$

$$\hat{f}_{Rt}(r) = \frac{\sum_{i,j} (1 - \varphi_i) \rho_{ij} I\left(\left|r_{ij} - r\right| < \Delta r/2\right)}{\Delta r \sum_{i,j} (1 - \varphi_i) \rho_{ij}}. \tag{40}$$

[43] The reconstruction results of the time lag and the location distributions are plotted in Figures 10 and 11. The time lag distributions for the background ancestors and the nonbackground ancestor are similar over all. However, we can still see that the events triggered by a triggered event decay slightly quicker than those triggered by a background event. For the simulated catalogue, there is no such difference between the triggered events and the background events. As to the distance distribution, the difference is near the zero distance, i.e., a background event triggers slightly more events in its close neighborhood than a triggered event.

8. Testing Dependence Between Components

[44] If each individual triggering process has no interaction with the others, six components need to be take into consideration in the mathematical formulation: the occurrence time t; the location (x, y) and magnitude M of the offspring; and the occurrence time t', location (x', y'), and magnitude M' of the ancestor event. Each of the first three components could depend on the other five components. In the space-time ETAS model, we simplified the above dependencies to $g(t|t')$, $f(x, y|x', y', M')$, and $J(M)$. In this section, we will check a number of such assumptions by using the reconstruction method.

8.1. Magnitude Distribution of Offspring Triggered by Different Magnitude Classes

[45] In the model formulation, we assume that the magnitude distribution does not depend on the magnitudes of their direct ancestors. To validate this assumption, we reconstruct the following empirical probability density functions:

$$\hat{J}(M|\mathcal{B}) = \frac{\sum_{i,j} \rho_{ij} I(M_i \in \mathcal{B}) I\left(\left|M_j - M\right| < \Delta M/2\right)}{\Delta M \sum_{i,j} \rho_{ij} I(M_i \in \mathcal{B})}, \tag{41}$$

where $\mathcal{B}$ is a set of magnitudes. Here we set $\mathcal{B}$ to be some interval of the magnitude ranges.

[46] The reconstruction results for both the JMA catalogue and the simulated catalogue are shown in Figure 12. For the JMA catalogue, the β values, if the exponential distribution is fitted to each of the curves, have a decreasing trend in the ancestor magnitude range from 4.2 to 6.1 and then become higher again after 6.2. The estimates of these β values are 2.3289, 1.9695, 1.6549, 1.4740, and 1.6090 for the ancestor magnitude intervals [4.2, 4.6], [4.7, 5.1], [5.2, 5.6], [5.6, 6.1], and [6.2, $+\infty$), respectively. On the other hand, when the same procedure is applied to the simulated catalogue, there is no corresponding dependence on the ancestor magnitudes. These results show that the magnitude distribution of the offspring depends on the magnitude distribution of the direct ancestor, which is different from the model assumptions.

8.2. Are Triggered Events Diffusive?

[47] One interesting issue is whether the clustering processes are diffusive. The causes of the diffusion of aftershocks may include (1) the generation of higher order offspring and (2) the diffusion of the process of generating direct offspring. The first aspect is already implemented inside the space-time ETAS model. The second aspect concerns the dependence between the occurrence times and locations of direct offspring, which is not included in

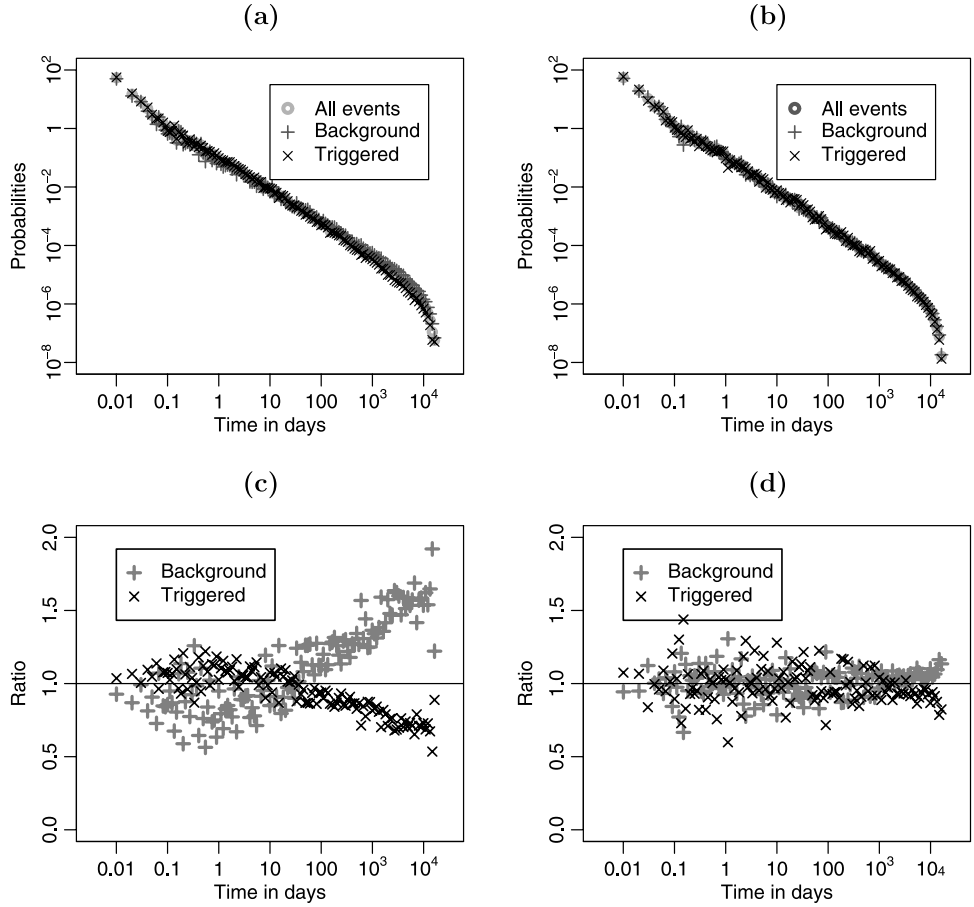


Figure 10. Reconstructed time distributions of offspring, $\hat{g}_b(t)$ in equation (37) of the background events and $\hat{g}_t(t)$ in equation (39) of the triggered events for (a) the JMA catalogue and (b)the simulated catalogue. The ratios $\hat{g}_b(t)/\hat{g}(t)$ (gray pluses) and $\hat{g}_t(t)/\hat{g}(t)$ (black asterisks) for (c) the JMA catalogue and (d) the simulated catalogue, where $\hat{g}(t)$ is defined by equation (24).

the model. To test the existence of such dependence, we construct the following empirical distribution of the locations conditional on a certain set of occurrence times,

$$\hat{f}(r|\mathcal{T}) = \frac{\sum_{t_{ij}\in\mathcal{T}\wedge|r_{ij}<r|<\Delta r}\rho_{ij}}{\Delta r\sum_{t_{ij}\in\mathcal{T}}\rho_{ij}}, \tag{42}$$

where r is the standardized distance defined by equation (25) and $\mathcal{T}$ is a small time interval.

[48] For convenience in comparison, we plot the image of the ratio

$$F(r,t) = \frac{\Delta t\hat{f}[r|(t-\Delta t, t+\Delta t)]}{f_R(r)\int_{t-\Delta t}^{t+\Delta t} g(s)ds} \tag{43}$$

in Figure 13 to detect whether each individual triggering process is diffusive, where $g(t)$ and $f_R(r)$ are defined in equations (5) and (27), respectively, with the parameters from fitting model II to the JMA catalogue. For the time less than 0.1 day, the image in Figure 13a seems to show a pattern of diffusion. This may be caused by the low detection rate of earthquakes immediately after the main shocks. For the time more than 0.1 day, $F(r, t)$ is close to a constant 1, indicating that there is no dependence between the occurrence times and locations. That is to say, diffusion of each individual triggering process, if it exists, is very weak. *Helmstetter et al.* [2003] obtained a similar conclusion from an analysis of 21 aftershock sequences in California. This also implies that after a large earthquake, the variation of the stress field due to stress diffusion is slow and rarely triggers seismicity.

8.3. Distributions of Offspring Locations From Different Magnitude Classes

[49] Because of the historical reasons mentioned in section 2, we model the distribution of locations of the direct offspring from an earthquake as an inverse power one with a scaling factor associated with the ancestor's magnitude, i.e., $D^2e^{\alpha(M-M_C)}$. Immediate questions for such a choice are as follows: (1) Is this scaling factor necessary? That is, can we use a constant D_0^2 to replace the scaling factor $D^2e^{\alpha(M-M_C)}$ in the model? (2) Is it necessary to link the scaling factor to the triggering ability, or should we introduce a new parameter γ instead of α for the scaling factor?

[50] To answer the above questions, for a small interval $\mathcal{M}$ of magnitudes, we select the pairs $\{(i, j)\}$ such that $M_i \in \mathcal{M}$ and then estimate the scaling factor $D^2_{\mathcal{M}}$ for $\mathcal{M}$, in a way

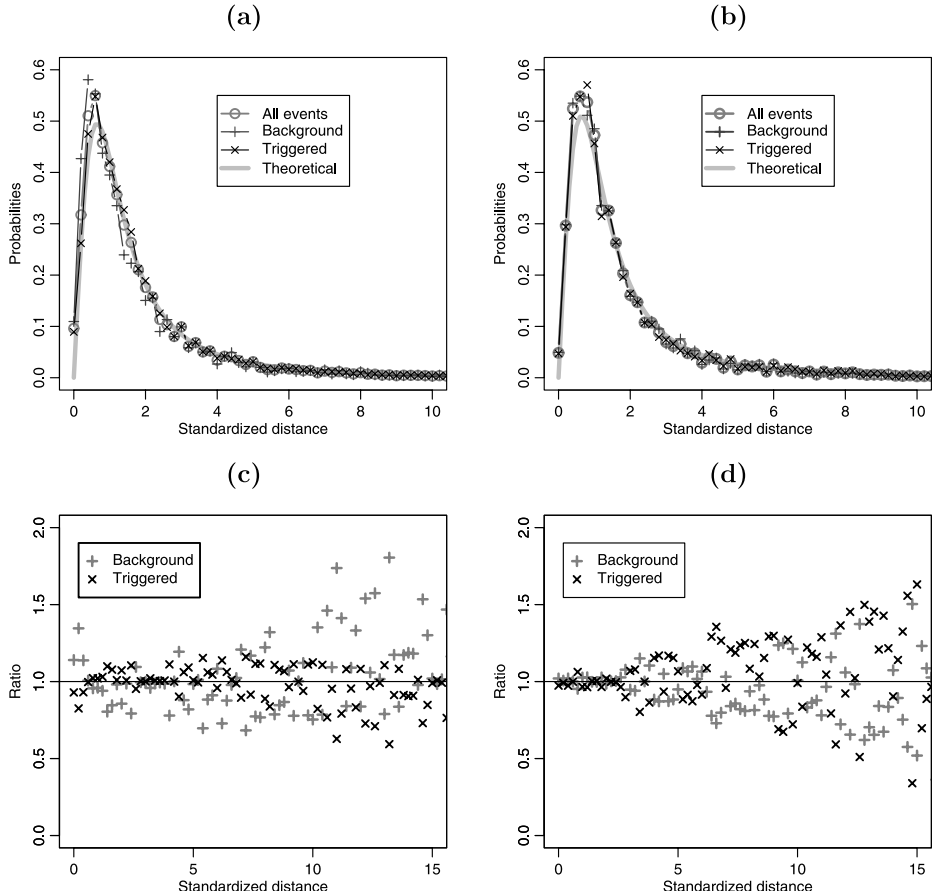


Figure 11. Reconstructed offspring distance distributions, $\hat{f}_{Rb}(r)$ in equation (38) of the background events and $\hat{f}_{Rt}(r)$ in equation (40) of the triggered events for (a) the JMA catalogue and (b) the simulated catalogue. The ratios $\hat{f}_{Rb}(r)/\hat{f}_R(r)$ (gray pluses) and $\hat{f}_{Rt}(r)/\hat{f}_R(r)$ (black asterisks) for (c) the JMA catalogue and (d) the simulated catalogue, where $\hat{f}_R(r)$ is defined by equation (28).

similar to the estimation of the parameters in mixture models by using the expectation-maximization algorithm [see, e.g., *Eggermont and LaRiccia*, 2001, section 2.4]. Given ρ_{ij}, consider the following pseudo log likelihood function:

$$\log L = \sum_{M_i \in \mathcal{M}} \sum_j \rho_{ij} \log \left[\frac{2(q-1) D^{2(q-1)} R_{ij}}{\left(R_{ij}^2 + D^2 \right)^q} \right]. \qquad (44)$$

To maximize it, let

$$\left. \frac{\partial \log L}{\partial (D^2)} \right|_{D^2 = D^2_{\mathcal{M}}} = 0,$$

i.e.,

$$\frac{q-1}{D^2_{\mathcal{M}}} \sum_{M_i \in \mathcal{M}} \sum_j \rho_{ij} - q \sum_{M_i \in \mathcal{M}} \sum_j \frac{\rho_{ij}}{R_{ij}^2 + D^2_{\mathcal{M}}} = 0. \qquad (45)$$

Thus we can construct the following iteration to solve the above equation

$$\left[D^2_{\mathcal{M}} \right]^{(n+1)} = \frac{(q-1) \sum_{M_i \in \mathcal{M}} \sum_j \rho_{ij}}{q \sum_{M_i \in \mathcal{M}} \sum_j \frac{\rho_{ij}}{R_{ij}^2 + \left[D^2_{\mathcal{M}} \right]^{(n)}}}. \qquad (46)$$

Figure 14 shows the values of $D^2_{\mathcal{M}}$ against the magnitude classes. We can see that values of $D^2_{\mathcal{M}}$ have a slope different from $\kappa(M)$, which is 0.5008 by the least square fit. Thus it is not suitable to use $\kappa(M)$ as the scaling factor; a better choice is to introduce another parameter γ as the coefficient in the exponential part.

9. Discussion

9.1. Stochastic Reconstruction and Model Selection

[51] Needless to say, the stochastic reconstruction method proposed in this paper has some similarities to and also some differences from model selection procedures. A model selection criterion can be applied for more general forms of models, and it gives a quantitative comparison between the models fitted to the same data. Stochastic reconstruction is only suitable for some branching models, such as the ETAS model, and neither it is a quantitative rule. The advantage of the stochastic reconstruction method is that it gives us a visual impression of the goodness of fit for each component in the formulation of the model but not a number showing the overall difference between the models. By using this method, we can still find the good points of a model whose overall fit may be worse than others. Picking up the good points of

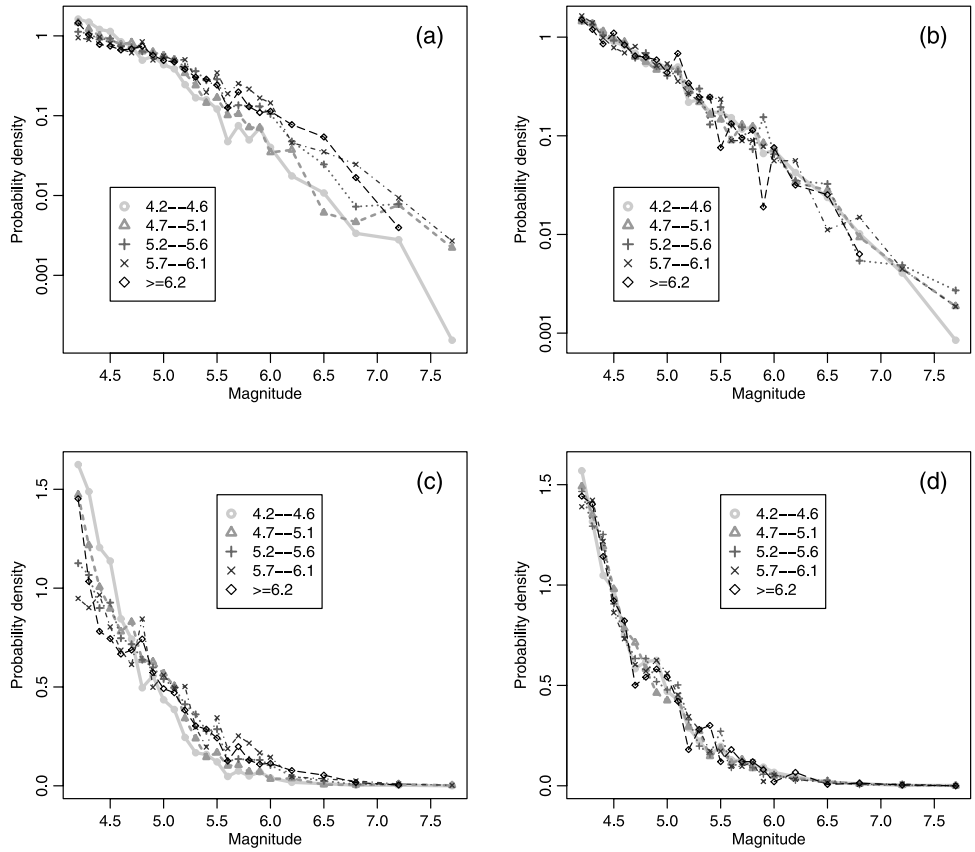


Figure 12. Reconstruction of the magnitude distributions, $\hat{J}(M|\mathcal{B})$ in equation (41) of the events triggered by events from different magnitude classes $\mathcal{B}$ (listed in the legends) for (a) the JMA catalogue and (b) the simulated catalogue. (c) and (d) Linear-scale versions of Figures 12a and 12b, respectively.

known models can help us to propose a better model without much difficulty.

9.2. Improper Model and Improvement

[52] One may argue that the reconstruction output is unreliable if the model is not properly fitted to the data. This may be true if the model diverges too much from the data. When applying the stochastic reconstruction method, the results are something between what are described by the model and the realities in the data. If the reconstruction results are much different from the model, it is necessary to propose a new model according to the reconstruction output and then do the reconstruction procedures again with the new model. We can repeat the above procedures many times in order to find a model good enough for describing the data. In this paper, we have shown that the

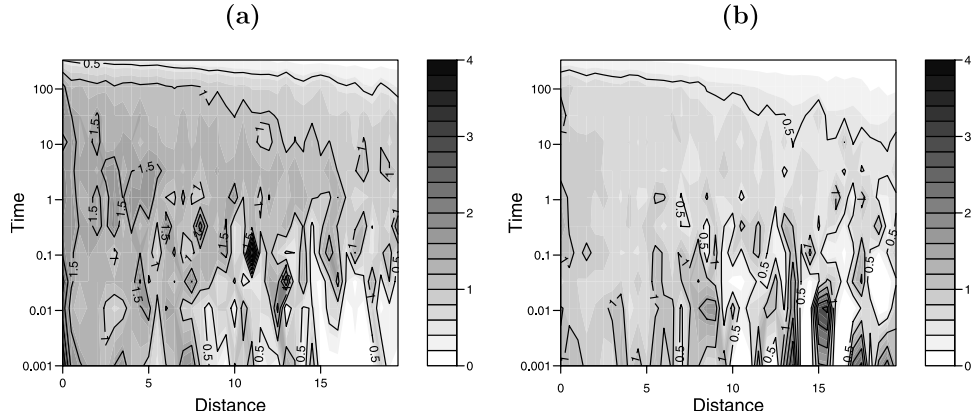


Figure 13. Images of $F(r, t)$ (see equation (43)) for (a) the JMA catalogue and (b) the simulated catalogue. The units are days for the vertical axis and degrees for the horizontal axis.

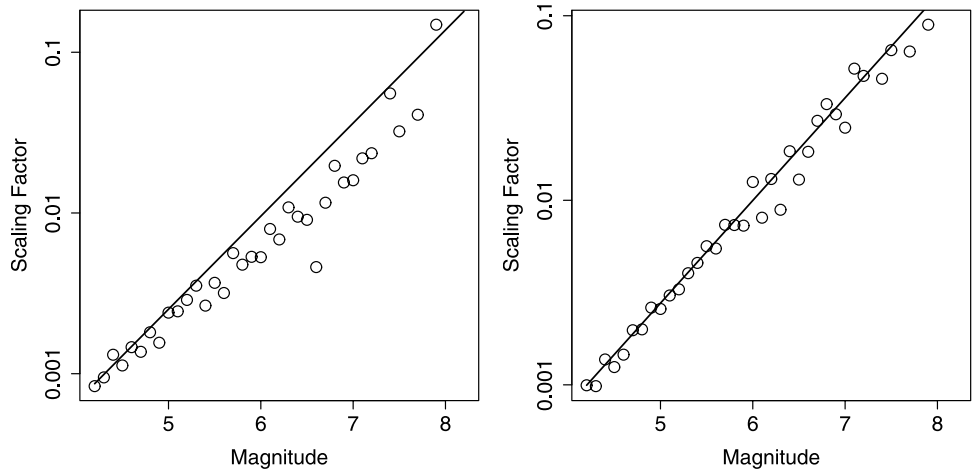


Figure 14. Reestimated $D^2_{\mathcal{M}}$ for (a) the JMA catalogue and (b) the simulated catalogue. Theoretical fitting curves, $D^2 e^{\alpha(M-M_C)}$, are represented by the straight lines.

ETAS model is good enough to be used as a first-order approximation, which ensures the reliability of the reconstruction results.

9.3. Checking Results by Simulation

[53] We only show one example of simulation in the comparison between the reconstruction results. Is one simulation enough for verifying the results? For the case of a low dimensional random variable, one simulation is, of course, insufficient. However, for the ETAS model, when it is stationary and ergodic, a simulation of a long enough time can cover all of its properties because the law of the large number. In this paper, the critical parameter, $\varrho < 1$, implies that the ETAS model fitted the JMA catalog is stationary and ergodic. There is a total of about 9000 events in the simulated catalog, which is a big enough number. The reasons explain why we do not need more simulations.

9.4. Future Research

[54] There can be many studies associated with the stochastic reconstruction method proposed above. First, the reconstruction results directly leave us a task of proposing a new model to handle the features not included the ETAS model and studying the seismicity with the new model. For example, the new model should consider the difference between triggering abilities of the background events and the triggered events or cut off the direct link between the spatial scaling factors and the triggering abilities.

[55] As mentioned by *Zhuang et al.* [2002], foreshocks are much better evaluated by the stochastic declustering method or the stochastic reconstruction method. It is important to find whether foreshocks produce main shocks in a different way from the one way that main shocks produce aftershocks. If different, what are the differences between them? Such a study will help us to better understand foreshocks.

[56] In this paper, we did not consider the confidence level for the difference between the reconstruction results and the function fitted from the overall data, but we made a judgment in a very rough way. A quantitative rule is necessary for future studies.

[57] An important feature of the estimation procedure is using a combination of a nonparametric method to estimate the background intensity and a parametric method to estimate the branching structure. Is there a general and reasonable criterion for selecting the best one among several models of this type that are fitted to the same data?

10. Conclusions

[58] Using the reconstruction techniques introduced by this paper, we analyzed the clustering features of earthquake occurrences. Even though the reconstruction is based on the ETAS model, we find several discrepancies of earthquake clustering from the model assumptions, as outlined below.

[59] 1. Even though not perfect, the ETAS model can describe the clustering phenomena sufficiently to be used as the reference model for analyzing the clustering features of earthquakes by using the stochastic reconstruction method.

[60] 2. The magnitude distributions are different for the background events and the triggered events: both can be described by the Gutenberg-Richter law but with different b values. The background events have a higher b value. These different b values may indicate the different inhomogeneity of the stress field before and during the burst of the clusters.

[61] 3. The time lag distribution between an triggered event and its direct ancestor can be well fitted by Omori's law. However, the occurrence frequencies of the events triggered by a triggered event decay slightly more quickly than those triggered by a background event.

[62] 4. The locations of triggered events are better represented by an inverse power law than by a normal density, and there is no significant difference in the location distributions of the events triggered by a background event and the events by a triggered events.

[63] 5. A background event triggers fewer events than a triggered event of the same magnitude.

[64] 6. The magnitude of a triggered event depends on the magnitude of its direct ancestor.

[65] 7. We found no evidence suggesting that the diffusion of aftershocks is due to the existence of diffusion in each individual triggering process. In other words, the diffusion of aftershocks is caused by the

cascade or superposition of these individual triggering processes.

[66] 8. The spatial scaling factor for the locations of the event triggered by an ancestor is not proportional to the triggering ability of this ancestor. However, the reconstruction results suggest that the scaling factor follows a different exponential law from the one for the triggering abilities.

[67] In summary, by using the stochastic reconstruction method, we can tackle the difficulties caused by the overlapping of clusters and better visualize and quantify the clustering features.

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